

STOR 320: Methods for Data Science

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Final Paper Group 6

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Introduction

Have you ever wondered if philanthropy in the form of item donation actually has a positive effect? What about if those items were medical? What if you had all the money in the world and you single-handedly wanted to prevent or stop a pandemic? Well, this study hopes to explore that. The two questions we intend to answer are: which donation—among masks, N95 masks, COVID-19 test kits, ventilators, and various dollar-valued donations that China provided to Latin American countries—was the most effective at reducing cases during the COVID-19 pandemic. Additionally, we intend to analyze the various donation strategies among the different Latin American countries deployed by donors in China to gain a better understanding of how said donations were dispersed.

These questions are important to consider because donations are not free, and when a particular government or company is looking to donate to foreign countries, it would be helpful to know which donation item would most positively affect the receiving country. It would also be useful to know when to donate or to identify common donation strategies to parallel.

What is most interesting about this, though, is the political and diplomatic implications. China was not the only country donating to Latin American countries, but they were the first. Analyzing and identifying their most common strategy could reveal intent and help set it in stark contrast to other countries and their intents with their donations.

Most importantly, COVID-19 had global impact, and hopefully through this study we can reveal how to mitigate pandemic outcomes globally through foreign aid most effectively.

Data

This study integrates two major datasets to examine the relationship between international COVID-19 donations and epidemic results in Latin American and Caribbean countries. The first dataset comes from the World Health Organization (WHO), which contains daily COVID-19 statistics from 237 countries from January 2020 to July 2022, including new cases, cumulative cases, new deaths, and cumulative deaths. The second dataset is a “mask diplomacy dataset” compiled by researchers who study the distribution of epidemic aid materials, which records donations shipped to Latin American countries, including the name of the donor, the receiving country, the date of donation, and specific items (such as masks, N95 masks, COVID-19 test kits, and ventilators), as well as the quantity, dollar value, and other detailed information.

Our analysis focuses on 33 countries in the Americas (World Health Organization Regional Office for the Americas), spanning from January 2020 to December 2020. This period covers the critical stages of early response to the epidemic. International assistance is most concentrated during this time, and the growth trajectory of cases and deaths varies significantly from country to country.

Data Cleaning and Merging

The data preparation process includes several key steps. First, we standardized the country names in the two datasets, such as mapping “Venezuela” to “Venezuela (Bolivar Republic)” and mapping “Bolivia” to “Bolivia (multi-ethnic country)” to ensure the accuracy of data consolidation. We screened the dataset from the World Health Organization and retained only the countries in the collection of donation data, and finally obtained 30,358 daily observations from 33 countries/regions. Because of heterogeneity in case rates and death rates between countries, we will do country-by-country modeling and analysis instead of aggregate analysis.

In order to reduce the inherent noise of daily data, we will summarize the observation data to the weekly level. For the weekly combination of each country, we calculated the average percentage change in the number of cases and deaths, as well as the total amount of all donations. After weekly summarization, we obtained 1,716 observations (33 countries/regions $\times$ 52 weeks). We also calculated a derivative variable called “Unaccounted_USD” by subtracting the estimated market value of the recorded items (0.40 USD per mask, 2.00 USD per N95 mask, 6.00 USD per test kit, and \$45,000 per ventilator), reflecting the monetary value of other or unrecorded aid.

Variable Descriptions

The multicollinearity test using the variance inflation factor (VIF) confirms that the VIF value of all predictors used is lower than 3, indicating that there is no significant multicollinearity problem, so there was no need to use dimension reduction techniques such as principal component analysis when we built our models.

Table 1 lists the key variables used in our analysis. Our result variables are Percent_increase_cases and Percent_increase_deaths, which measure the average change rate of the number of new cases and deaths per week, respectively. Predictive variables include five types of donations: ordinary masks, N95 masks, COVID-19 tests, ventilators and undeclared monetary donations (undeclared US dollars).

Variable	Type	Description	Range
Country	Categorical	Latin American/Caribbean nation name	33 countries
Percent_increase_cases	Continuous	Weekly average percent change in new COVID-19 cases	-1.72 to 0.49
Percent_increase_deaths	Continuous	Weekly average percent change in new COVID-19 deaths	-1.0 to 0.74
masks	Count	Number of standard surgical masks donated per week	0 to 2,000,000+
masks_n95	Count	Number of N95 respirator masks donated per week	0 to 500,000+
tests	Count	Number of COVID-19 diagnostic tests donated per week	0 to 100,000+
ventilators	Count	Number of mechanical ventilators donated per week	0 to 50+
Unaccounted_USD	Continuous	Monetary donations minus estimated value of documented items (USD)	0 to \$10,000,000+

Table 1: Variable Descriptions

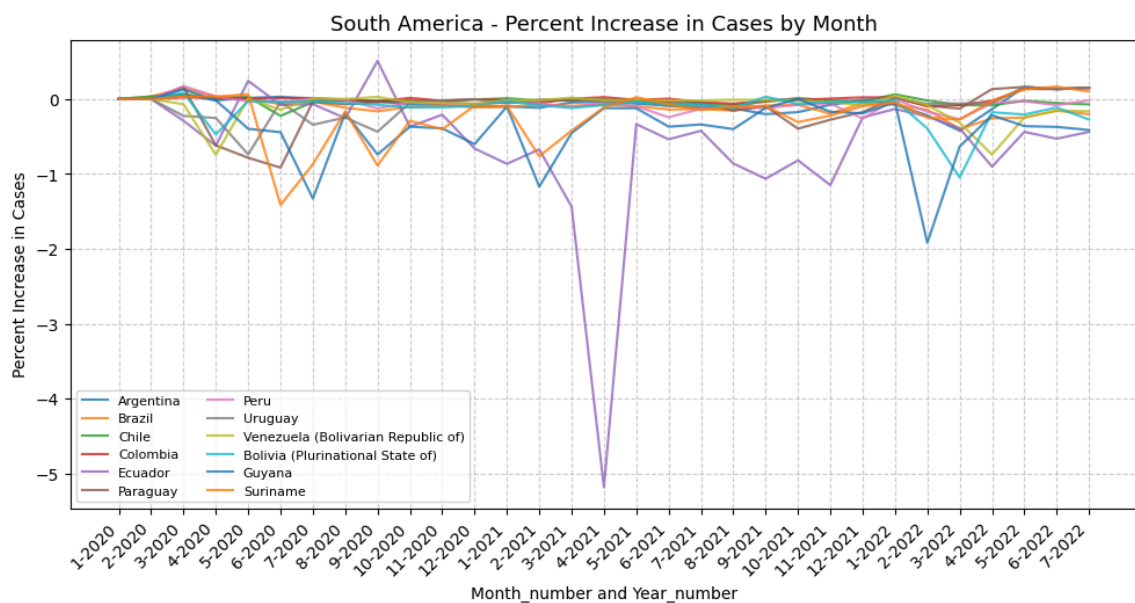


Figure 1: Distribution of COVID-19 Case Growth Rates by Region

Figure 1 shows the varying average case growths over countries in South America per month. The heterogeneity in the case rates between countries seen is why we chose to do a country-by-country analysis instead of an aggregate analysis.

Results

Question 1: Which Donation Type Was Most Influential/Effective?

In order to determine which donation type can best predict the changes in the results of the COVID-19 epidemic, we adopted two modeling methods: random forest regression was used to identify the most influential donation types, and the distributed lag model (compared to the OLS baseline model) was used to choose ideal lags and then use those ideal lags to determine which donation item was the most effective.

Random Forest Analysis

First, we applied the random forest regression model to each country/region, adopted a 20:80 training set/test set ratio, and used the parameters $n_estimators = 1000$, $max_depth = 12$, $max_features = 'sqrt'$, and $min_samples_leaf = 3$. The model uses five donation categories as features to predict the percentage change in the number of cases and deaths per week. The model then calculates a relative importance score for each donation type to influence case and death rates.

There is significant heterogeneity in how well the model performed on the countries. In terms of case prediction, only 13 out of 33 countries obtained positive out-of-sample R^2 (OSR^2) and any importance greater than 0, meaning that the model did not work on 20 of the countries for case rates. In terms of death prediction, only 22 out of 33 countries obtained positive OSR^2 and any importance greater than 0, meaning that the model did not work on 11 of the countries for death rates.

In Figure 2, you can see that the most influential type of donation for both case and death rates was "Unaccounted_USD," followed by "Masks," meaning that possibly other donation items than ordinary masks, N95 masks, COVID-19 tests, and ventilators were more influential.

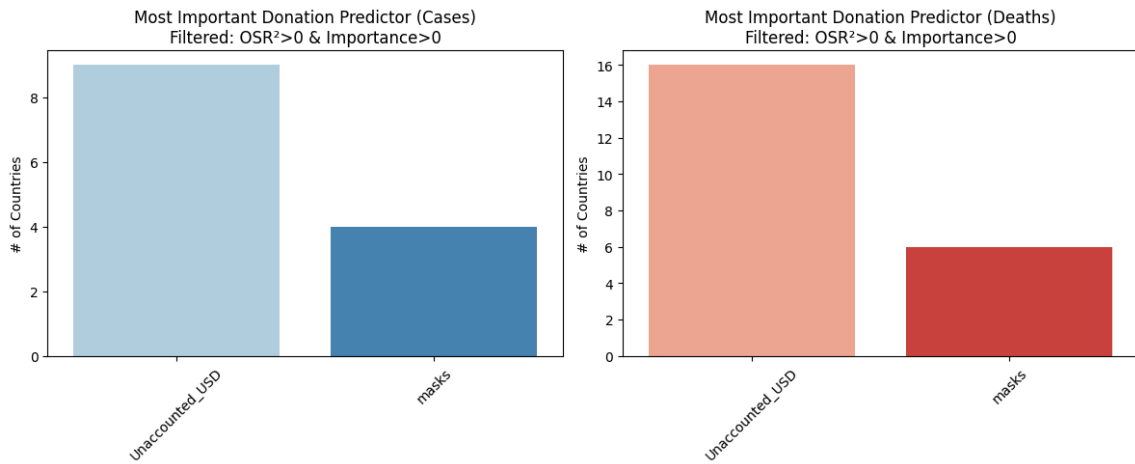


Figure 2: Frequency of Most Influential Donation Types Across Countries

Distributed Lag Model with Optimal Timing

In order to actually measure the effectiveness of each donation type, we used an epidemiological method called a distributed lag model to test the lag time from -10 weeks (case rates before donations) to $+15$ weeks (case rates after donations). For each country/region, we chose the lag that maximizes R^2 after adjusting the training data, and then used OSR^2 to evaluate the performance of the model on reserved test data. Compared with a benchmark OLS model at lag 0, this method significantly improves the fit of the model. The average OSR^2 of case prediction increased from -0.065 (reference OLS model) to 0.049 (distributed lag model), and the average OSR^2 of death prediction increased from -0.11 (reference OLS model) to -0.0050 (distributed lag model).

Knowing our model works better, we then computed effectiveness by first looking at negative coefficients and then only countries with positive lag; that way, when a negative effect occurs because of a donation, we know the donation was most likely before that negative effect. We then standardized those coefficients, and we then weighed coefficients based on whether their lag was more positive (to avoid the results reflecting reactive behavior), sample size, and model predictive skill (OSR^2). For each donation type, we summed the coefficients from each country for that particular item (if it was not filtered out before). The results, which can be seen in Figure 3, showed that ventilators were the most effective at decreasing case rates and tests were the most effective at decreasing death rates.

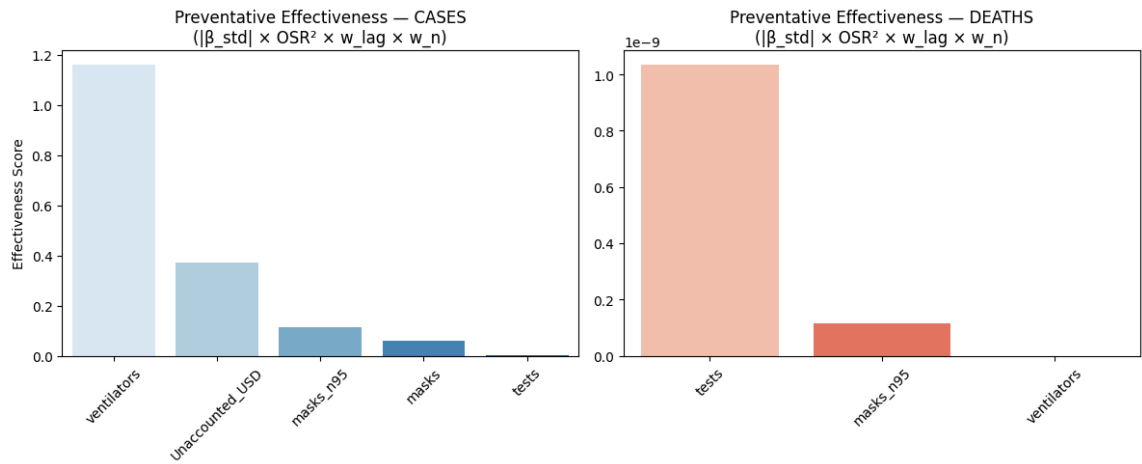


Figure 3: Aggregate Effectiveness Score For Each Country

Question 2: *What Was the Donation Strategy for Each Country?*

In order to describe the donation mode of each country, we divide each country into two categories: "preventive" and "reactive" according to the optimal lag time determined in the distributed lag model. This classification reflects the timing strategies of international donors to provide assistance to countries. Within the positive lag time (i.e., the donation time is more than two weeks earlier than the case change), the countries with the strongest correlation between donations and results are classified as preventive/proactive, indicating that donors provided assistance before the large-scale outbreak of the epidemic. Countries with an optimal delay time equal to or less than +1 week are classified as reactive, indicating that donors provided assistance after the outbreak.

Our classification results seen in Figure 4 show that 12 countries received mostly proactive donations, 11 received mostly reactive donations for cases, 1 received mostly reactive donations for deaths, and the model failed to classify 7 countries.

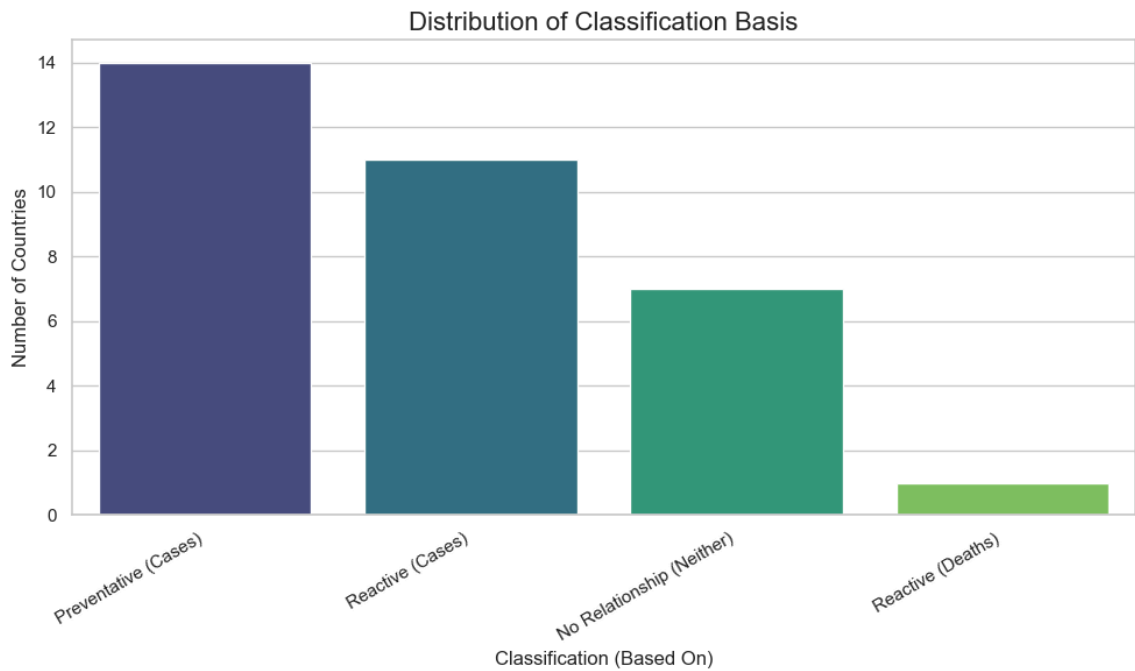


Figure 4: Distribution of Donation Strategies Across Countries

Lagged LASSO Model

In order to verify our classification and determine the specific donation items that drive each strategy, we applied LASSO regression and extended the lag window from -21 to +21 weeks. LASSO's L1 penalty term naturally selects features by narrowing the less important coefficients to zero, so that we can determine which donation-lag combinations are the most predictive for each country/region. Once we had a set of important features, we calculated a reactivity and proactivity score for each country based on the coefficients for those features by summing them according to their sign. Reactivity is based on the negative coefficients, and proactivity is based on the positive coefficients. We then classified donation strategy based on which score was greater for each country. This model worked much better because, regardless of how small the scores were in each country, every country was classified.

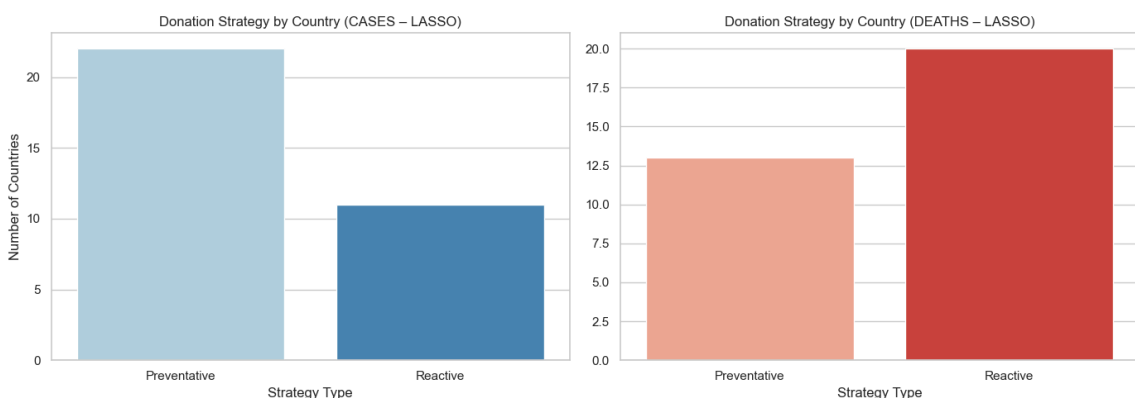


Figure 5: Preventative vs Reactive Country Counts (LASSO – Deaths and Cases)

In Figure 4 we see that the donation strategies were proactive in more countries for cases and reactive in more countries for deaths suggesting that deaths may have been more of a concern to those who were making donations.

Conclusion

Our group built these models to answer the following questions: Which type of donation was most effective/influential on epidemic outcomes from January 2020 to December 2020, and what specific strategies characterized each country's reception of aid? Our analysis revealed that masks and unaccounted financial aid were considered the most influential predictors of weekly case change according to the random forest model. In the distributed lag model, ventilators were the most effective at reducing case rates, suggesting that mechanical respiratory support may have alleviated healthcare system strain enough to suppress broader transmission dynamics. Interestingly, testing kits showed the strongest effectiveness at reducing death rates. This suggests that detection of COVID-19 played a meaningful role in preventing mortality. For our second question, we found that most donation strategies for Latin countries were preventive for cases according to the distributed lag model. The Lagged Lasso Model found very similar results but also found that donations were mostly reactive to deaths.

Our results are important because optimizing the timing and type of donations sent can ensure that limited resources are used effectively. In epidemiology and public health, an important topic that is often discussed is policies for allocation of resources within hospitals. If another country donates a large quantity of masks instead of ventilators—keeping in mind that our models showed that ventilators were more effective at reducing cases—it is entirely possible that sick or dying people are turned away by hospitals that lack the more essential resource. Ventilators are commonly at the forefront of these discussions, and when a hospital does not have enough ventilators or other important resources, they become more stringent about what they allocate and to whom, possibly even slowing down diagnosis or, in most cases, treatment. That is why timing is important too. If a particular area has what they need in advance, they can better prepare for future rises in cases. Corroborating the results for each question, as well as keeping the importance of these questions in mind, proactive donations of ventilators could definitely be the most ideal strategy in similar efforts.

If any future work is done with this, it would be interesting to look more in-depth at the regional level or examine the strategies of specific donor types (i.e., government, companies, or individuals). America provided similar support to Latin countries during COVID-19, so if there were data on American donations as well, they could also be incorporated into future analyses. If that was explored, we could observe diplomatic intent, since both countries had

different goals of diplomacy in Latin countries. Providing a more in-depth analysis of an intent–effectiveness relationship would be valuable. There are also many more factors that could have confounded our results, and finding a way to incorporate them would also be interesting (i.e., donation disbursement, lockdown policy, etc.).